SECTION 2 · SUBSECTION 2.1
PERCEPTION AND DECISION CORE

2.1.1 INTRODUCTION.

The agent plays POKEMON BLUE by understanding each room, not by replaying scripts. It sees every room twice: the RAM reader reports it as a 10 by 9 tile grid from emulator memory, and a vision model reports the same screen as a character grid. The two pictures are compared cell by cell every frame; every disagreement is a finding (figure 2.1-1).

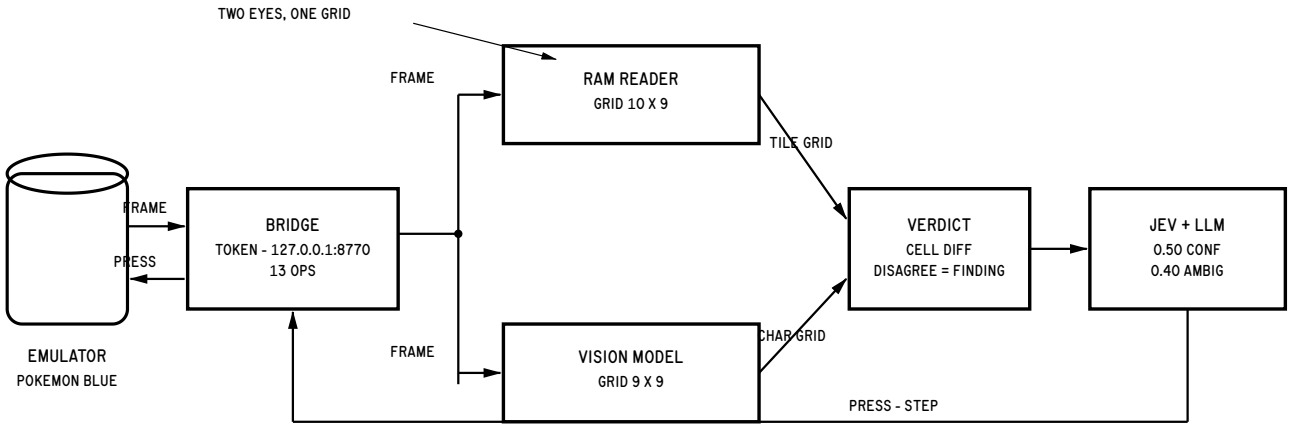



Figure 2.1-1. Two Perception Channels Converging On One Verdict

2.1.2 FUNCTIONAL DESCRIPTION.

Each cycle the bridge returns one frame and the RAM reader returns the room tile grid. The vision model describes the same frame as a character grid using the floor, wall, grass, water, doorway and person symbols. The verdict compares the grids cell by cell and reports every disagreement as a finding. JEV decides each cycle; below 0.50 confidence or above 0.40 ambiguity it hands the state, not the action, to the reasoning model (figure 2.1-2).

2.1.2.1 Two Grids, One Law.

The RAM reader renders 10 columns by 9 rows with the facing arrow at row 3; the vision module declares 9 by 9 with the player at row 4. Until one definition is shared, every verdict row is one row high and the comparison is wrong by construction (requirement R-001, row VISION-GRID-UNIFY).

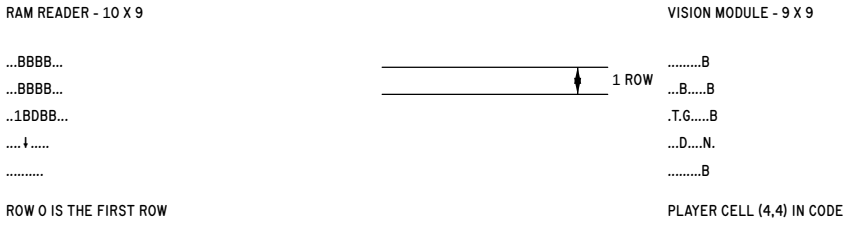



Figure 2.1-2. The Measured One-Row Offset Between The Two Grid Definitions

2.1.3 PERFORMANCE AND DESIGN DATA.

Table 2.1-1. Measured System Data

| Parameter | Value |
|---|---|
| Bridge listen | 127.0.0.1:8770, token required |
| RAM grid | 10 cols x 9 rows |
| Vision grid (code) | 9 x 9 - one row high |
| Player cell | RAM (4,3) - code (4,4) |
| Vision upscale | x6 LANCZOS, cap 1024 px |
| Handoff thresholds | 0.50 conf - 0.40 ambig |

**CAUTION**

TWO DRIVERS ON ONE BRIDGE FIGHT. SET LIVE OFF BEFORE AN AGENT RUN. THE TICKER ALSO DEGRADES THE VIDEO AFTER ABOUT 4,000 FRAMES.

| START A RUN | OPERATOR |
|---|---|
| 1 BRIDGE - verify health op | |
| 2 LIVE - set OFF | |
| 3 MODE - pick JEV/LLM/HYBRID | |
| √ LOG - mode echoed at run head | |
| 4 RUN - cron_runner --cycles 20 | |
| 5 VISION - ask now (25k tok) | |
| √ VERDICT - agree/conflict | |

AIP